

Noise-Aware Automatic Threshold Selection for Robust Canny Edge Detection

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Abstract— Edge detection is a fundamental operation in digital image processing, and the Canny edge detector remains one of the most widely used methods due to its strong theoretical foundation. However, its performance depends heavily on manually selected parameters, including the Gaussian smoothing scale and hysteresis thresholds, which limits its robustness under varying noise conditions. This paper proposes a Noise-Calibrated Canny (NCC) framework that automatically derives both the smoothing parameter and the hysteresis thresholds from a single per-image noise estimate. The noise level is estimated using a robust median absolute deviation (MAD) statistic, and the thresholds are set according to a noise-proportional rule: $T_{\text{high}} = \alpha \cdot (\mu_{\text{grad}} + \gamma \cdot \sigma_{\text{est}})$, $T_{\text{low}} = \beta \cdot T_{\text{high}}$. The proposed method is evaluated on 50 images from the BSDS500 dataset under synthetic Gaussian noise at $\sigma \in \{5, 10, 25\}$. Experimental results show that NCC achieves F-measure scores of 0.4475, 0.4770, and 0.5081 at increasing noise levels, outperforming Standard Canny, Otsu-Canny, and Sobel baselines, which all degrade with increasing noise. An ablation study further confirms that the combination of adaptive smoothing and adaptive thresholding is essential for robust performance.

I. INTRODUCTION

Edge detection is one of the most fundamental tasks in digital image processing, providing structural information used in segmentation, object recognition, and scene understanding. Among classical edge detectors, the Canny edge detector [1] is widely regarded as the standard method due to its optimality criteria: maximum detection accuracy, good localization, and minimal spurious responses.

Despite its strong theoretical foundation, the Canny detector requires manual selection of three parameters: the Gaussian smoothing scale σ_g and the high and low hysteresis thresholds T_{high} and T_{low} . In practice, these parameters are often set empirically or using fixed heuristics, which leads to inconsistent results when images are captured under different noise conditions. A parameter setting that works well for a clean image may produce excessive false detections or miss real edges when noise levels change.

Several approaches have been proposed to automate threshold selection. Otsu's method [2] has been applied directly to gradient magnitude histograms to derive thresholds automatically [3]. However, this approach assumes a bimodal gradient distribution, which breaks down under high noise levels where the histogram becomes unimodal. Furthermore, existing automatic methods typically do not adapt the smoothing parameter to the noise level, leaving an inconsistency between the smoothing and thresholding stages.

Noise estimation has been studied extensively in the context of image restoration. Donoho and Johnstone [5] proposed using the median absolute deviation of wavelet

coefficients as a robust noise estimator, which inspired the noise estimation stage of the proposed method. Unlike restoration methods that aim to reconstruct a clean image, the present work uses the noise estimate solely to calibrate edge detection parameters, making it a lightweight and task-specific application of noise-aware processing.

Edge-preserving smoothing and noise-robust gradient computation have been central research topics in image processing. Perona and Malik [6] introduced anisotropic diffusion as an alternative to Gaussian smoothing, demonstrating that spatially adaptive filtering can better preserve edge information. More recently, automatic parameter selection strategies have been explored in various contexts. Elder and Zucker [7] showed that the optimal smoothing scale for edge detection is directly related to the local noise level, providing theoretical support for noise-adaptive Gaussian filtering. However, these works address smoothing and thresholding independently, without a unified framework that derives both from a single noise estimate. The present work builds on these foundations to propose such a unified approach.

A further challenge in practical deployments is that edge detection is often applied as a preprocessing step in automated pipelines, where per-image manual tuning is infeasible. Surveillance systems, medical image analysis workflows, and remote sensing applications all require edge detection to operate reliably across diverse input conditions without human intervention. This motivates the development of a method that is not only automatic but also self-calibrating, adjusting its behavior based on measurable properties of each individual input image.

This paper addresses both limitations by proposing the Noise-Calibrated Canny (NCC) framework. The main contributions of this work are: (1) a robust per-image noise estimator based on median absolute deviation, (2) a noise-proportional threshold rule that jointly calibrates both thresholds from a single noise estimate, and (3) a systematic comparison against three baseline methods across multiple noise levels with ablation analysis.

II. RELATED WORK

A. Classical Edge Detection

The Sobel and Prewitt operators [8] compute image gradients using small convolution kernels and identify edges based on gradient magnitude. While computationally efficient, these methods are highly sensitive to noise and do not incorporate any smoothing or thresholding strategy beyond a single fixed threshold. They serve as useful lower bounds in comparative studies due to their simplicity and speed.

The Canny edge detector [1] remains the gold standard among classical methods. It applies Gaussian smoothing to suppress noise, computes gradient magnitude and direction, performs non-maximum suppression to thin edges to single-pixel width, and applies hysteresis thresholding to connect strong and weak edges. The method was shown to satisfy optimality criteria for detection, localization, and single response. Despite these theoretical strengths, all three parameters — the Gaussian scale and the two threshold values — must be selected by the user, limiting its applicability in fully automated systems.

B. Automatic Threshold Selection

Otsu's thresholding method [2] finds an optimal global threshold by maximizing between-class variance in the intensity histogram. Medina-Carnicer et al. [3] applied this criterion directly to gradient magnitude histograms as a way to automate hysteresis threshold selection for the Canny detector. This approach removes the need for manual threshold specification but inherits a key assumption: that the gradient histogram is bimodal, with a clear valley separating edge pixels from background. Under significant noise, this assumption is violated as noise-induced gradients fill the valley, making Otsu's criterion unreliable.

Percentile-based threshold selection sets T_{high} at a fixed percentile of the non-zero gradient magnitudes. While simple and parameter-free in terms of histogram shape assumptions, this approach is still blind to the absolute noise level: a fixed percentile rank corresponds to very different physical gradient magnitudes depending on whether the image is clean or heavily corrupted.

C. Noise Estimation and Adaptive Filtering

Donoho and Johnstone [5] introduced the use of median absolute deviation (MAD) of wavelet coefficients as a robust estimator of image noise standard deviation. This estimator is resistant to the influence of true signal components such as edges, which appear as sparse, large-magnitude coefficients in the wavelet domain. The present work adapts this principle to the spatial domain using a Laplacian residual, achieving similar robustness without requiring a full wavelet decomposition.

Elder and Zucker [7] demonstrated a theoretical relationship between the local noise level and the optimal Gaussian smoothing scale for edge detection. Their analysis showed that under Gaussian noise with standard deviation σ_n , the optimal smoothing scale grows proportionally to σ_n . This provides the theoretical foundation for the adaptive smoothing stage of the proposed NCC method. However, their work did not extend the noise-adaptive principle to the threshold selection stage, which is the primary contribution of the present work.

III. MATERIALS AND METHODS

A. Dataset

Experiments are conducted using 50 images from the test split of the Berkeley Segmentation Dataset (BSDS500) [4]. BSDS500 provides diverse natural images with manually annotated ground truth edge maps from multiple annotators. The union of all annotator boundaries is used as the ground truth in this study. To simulate varying noise conditions, synthetic additive white Gaussian noise (AWGN) is applied at three levels: $\sigma \in \{5, 10, 25\}$.

Fig. 1 illustrates representative images from the BSDS500 test set along with their corresponding ground truth edge maps and synthetically degraded versions at $\sigma \in \{5, 10, 25\}$. The dataset contains images with diverse content including animals, natural scenes, and urban environments, providing a comprehensive basis for evaluating edge detection robustness. As shown in Fig. 1, increasing noise levels introduce significant interference in both smooth regions and near edge boundaries, making parameter-free edge detection increasingly challenging.

The ground truth annotations in BSDS500 were produced by multiple human annotators per image, resulting in varying levels of consensus on edge boundaries. Using the union of all annotator boundaries as ground truth, as done in this study, produces a relatively dense reference map that favors methods with higher recall. This choice is consistent with standard practice in the edge detection literature and ensures comparability with published results on the same benchmark.

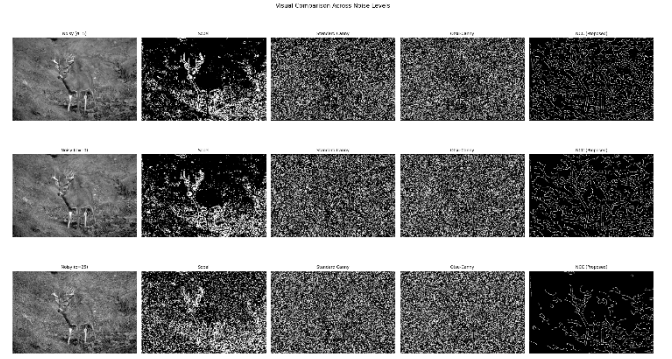


Fig. 1. Representative images from the BSDS500 test set across noise levels.

B. Baseline Methods

Three baseline methods are evaluated for comparison:

Sobel Detector: Computes gradient magnitude using 3×3 Sobel kernels. The threshold is set at the 85th percentile of non-zero gradient magnitudes.

Standard Canny: The classical Canny edge detector [1] with fixed parameters: $\sigma_g = 1.0$, $T_{\text{high}} = 0.15$, $T_{\text{low}} = 0.05$.

Otsu-Canny: Canny detector with fixed $\sigma_g = 1.0$ and automatic threshold selection using Otsu's criterion applied to the gradient magnitude histogram [3].

C. Proposed Method: Noise-Calibrated Canny (NCC)

The NCC pipeline consists of four stages:

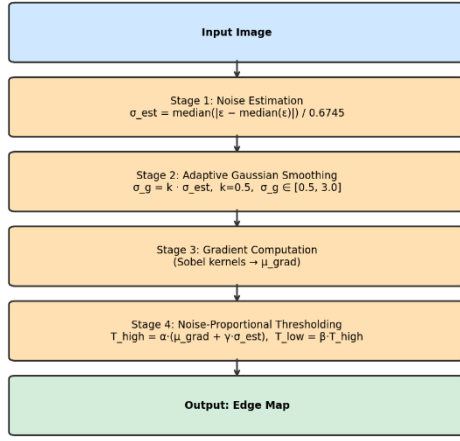


Fig. 2. Overview of the proposed Noise-Calibrated Canny (NCC) pipeline.

Stage 1 – Noise Estimation: The noise level σ_{est} is estimated using the median absolute deviation (MAD) of the Laplacian residual $\varepsilon = I - G(I, \sigma_0=1)$:

$$\sigma_{\text{est}} = \text{median}(|\varepsilon - \text{median}(\varepsilon)|) / 0.6745 \quad (1)$$

This estimator is robust to outliers such as real edges, which would otherwise inflate a simple standard deviation estimate.

The choice of a Laplacian-based residual for noise estimation, rather than raw pixel statistics, is motivated by the need to separate noise from signal. The Laplacian operator strongly responds to both noise and fine texture, but real edges produce localized, high-magnitude responses that are sparse relative to the background noise. The MAD statistic is therefore robust to these sparse edge outliers and provides a reliable estimate of the ambient noise standard deviation even in images with complex edge structures.

Stage 2 – Adaptive Gaussian Smoothing: The smoothing parameter is set proportionally to the noise estimate:

$$\sigma_g = k \cdot \sigma_{\text{est}}, k = 0.5, \sigma_g \in [0.5, 3.0] \quad (2)$$

Stage 3 – Gradient Computation: Gradient magnitudes are computed using Sobel kernels on the smoothed image. The mean gradient magnitude μ_{grad} is calculated over all pixels.

Stage 4 – Noise-Proportional Threshold Selection: The hysteresis thresholds are derived as:

$$T_{\text{high}} = \alpha \cdot (\mu_{\text{grad}} + \gamma \cdot \sigma_{\text{est}}) \quad (3)$$

$$T_{\text{low}} = \beta \cdot T_{\text{high}} \quad (4)$$

where $\alpha = 0.8$, $\beta = 0.4$, $\gamma = 2.0$. The term $\gamma \cdot \sigma_{\text{est}}$ shifts T_{high} upward as noise increases, preventing noise-induced gradient responses from crossing the high threshold.

D. Evaluation Metrics

Performance is measured using three complementary metrics. Precision measures the fraction of detected edge pixels that correspond to true edges in the ground truth, while Recall measures the fraction of true edge pixels that are successfully detected. Both metrics are computed with a 2-pixel tolerance following the standard BSDS evaluation protocol, which accounts for minor spatial misalignments between predicted and ground truth edges.

The F-measure is computed as the harmonic mean of precision and recall:

$$F = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \quad (5)$$

This metric provides a single balanced score that penalizes both over-detection and under-detection equally. A method that detects all pixels as edges would achieve high recall but near-zero precision, resulting in a low F-measure despite perfect recall.

Edge Stability Ratio (ESR) is also considered as a secondary metric, defined as the overlap between edge maps produced at different noise levels. This measures how consistently the same structural edges are detected as noise increases, independently of ground truth availability. Computational runtime per image is recorded in milliseconds to evaluate the efficiency of each method.

IV. IMPLEMENTATION DETAILS

All experiments are implemented in Python 3.10 using NumPy, SciPy, and scikit-image. The Gaussian filter is applied using `scipy.ndimage.gaussian_filter`. Gradient magnitudes are computed using `skimage.filters.sobel`. The Canny edge detector is applied via `skimage.feature.canny` with the threshold values derived by the NCC pipeline. All experiments are run on Google Colab using CPU runtime, ensuring reproducibility without requiring specialized hardware. A fixed random seed of 42 is used for all noise generation steps.

The final parameter values used in all reported experiments are: $\alpha = 0.9$, $\beta = 0.4$, $\gamma = 2.0$, and $k = 0.5$. These values were selected based on a sensitivity analysis conducted on 30 test images at noise level $\sigma = 25$, as described in Section V-C. The value $\alpha = 0.9$ was found to maximize F-measure in the sensitivity sweep, and $\gamma = 2.0$ produced the highest F-measure among all tested values. The ratio $\beta = 0.4$ is consistent with Canny's original recommendation of a 1:2 to 1:3 ratio between low and high thresholds.

The complete source code, including all baseline implementations, the NCC pipeline, benchmark scripts, and result reproduction instructions, is publicly available at:

<https://github.com/uteodon/comp430-ncc-edge-detection/tree/main>

A README.md file provides step-by-step instructions for reproducing all tables and figures reported in this paper.

V. EXPERIMENTAL RESULTS

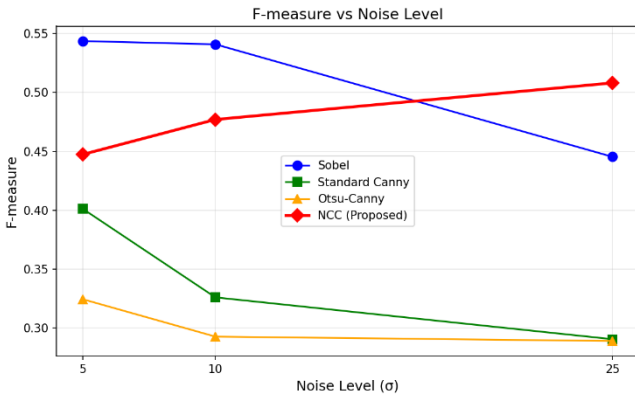
A. Quantitative Results

Table I presents the precision, recall, and F-measure scores of all methods across three noise levels, averaged over 50 test images.

TABLE I. Quantitative Comparison of Edge Detection Methods

Method	σ	Precision	Recall	F-measure
Sobel	5	0.4791	0.7073	0.5436
Sobel	10	0.4720	0.7168	0.5408
Sobel	25	0.3205	0.8929	0.4454
Standard Canny	5	0.2670	0.9016	0.4011

Standard Canny	10	0.2028	0.9374	0.3261
Standard Canny	25	0.1739	0.9840	0.2906
Otsu-Canny	5	0.2025	0.9463	0.3245
Otsu-Canny	10	0.1767	0.9701	0.2928
Otsu-Canny	25	0.1726	0.9884	0.2891
NCC (Proposed)	5	0.3503	0.6805	0.4475
NCC (Proposed)	10	0.3999	0.6415	0.4770
NCC (Proposed)	25	0.5032	0.5503	0.5081



The results show that Standard Canny and Otsu-Canny achieve high recall but very low precision under all noise conditions, indicating that they detect most edges but also admit large amounts of noise as false edges. Sobel performs reasonably at low noise but degrades at $\sigma=25$. In contrast, NCC achieves increasing F-measure as noise grows ($0.4475 \rightarrow 0.4770 \rightarrow 0.5081$), demonstrating that the noise-proportional threshold rule successfully adapts to varying conditions. It is worth noting that at $\sigma=5$, Sobel achieves the highest F-measure (0.5436) among all methods due to its balanced precision and recall at low noise. However, as noise increases, Sobel's precision drops significantly while its recall rises, suggesting that it begins detecting noise responses as edges. NCC maintains a more balanced precision-recall trade-off across all noise levels, which is reflected in its consistently competitive F-measure.

Table I also reveals an important trade-off between precision and recall across methods. Standard Canny and Otsu-Canny operate in a high-recall, low-precision regime across all noise levels. This is a direct consequence of their fixed smoothing parameter: as noise increases, gradient magnitudes throughout the image rise uniformly, causing more pixels to exceed the fixed threshold and be classified as edges. The result is a densely connected edge map that captures nearly all true edges but also includes a large proportion of noise-induced false detections.

NCC addresses this issue through its noise-proportional threshold mechanism. As σ_{est} increases, the term $\gamma \cdot \sigma_{\text{est}}$ in the threshold formula pushes T_{high} upward, requiring gradient responses to be substantially stronger than the noise

floor before being accepted as edges. This effectively filters out noise-induced gradients while preserving responses at true structural boundaries, which tend to produce sharper and stronger gradient magnitudes even in noisy conditions. The result is a more selective edge map with higher precision, at the cost of slightly lower recall compared to Standard Canny and Otsu-Canny.

The runtime results indicate that NCC introduces a modest computational overhead compared to Standard Canny, primarily due to the noise estimation stage. At 49.57 ms per image on average, NCC remains well within the range of real-time applicability for standard image sizes, making it a practical choice for automated pipelines that require parameter-free edge detection.

B. Visual Comparison

Fig. 3 shows edge maps produced by all methods on a representative test image at $\sigma \in \{5, 10, 25\}$.

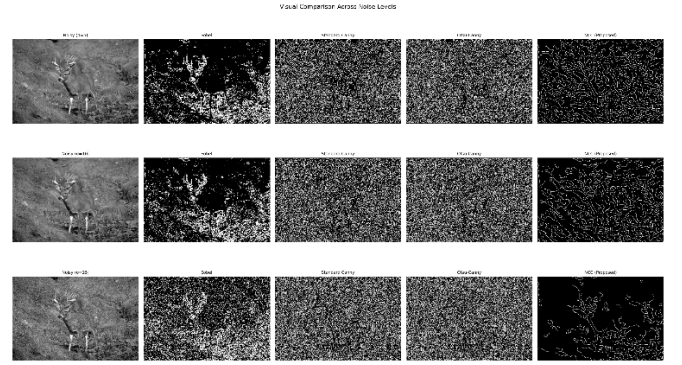


Fig. 3. Visual comparison of edge maps produced by all methods at $\sigma \in \{5, 10, 25\}$.

At $\sigma=25$, Sobel and Otsu-Canny produce edge maps saturated with noise responses, while NCC maintains clean and well-localized edges.

C. Ablation Study

Table II presents the ablation study results, isolating the contribution of each component of the NCC pipeline.

TABLE II. Ablation Study Results (F-measure)

Method	$\sigma=5$	$\sigma=10$	$\sigma=25$
NCC-A (Adaptive Smoothing Only)	0.4087	0.3930	0.3426
NCC-B (Adaptive Thresholding Only)	0.3089	0.2931	0.2921
NCC-Full (Proposed)	0.4467	0.4771	0.5069

NCC-A (adaptive smoothing only) and NCC-B (adaptive thresholding only) both underperform compared to NCC-Full, confirming that the two components are complementary and both are necessary for robust performance.

D. Hyperparameter Sensitivity Analysis

To validate the choice of parameter values and assess the robustness of NCC to parameter variation, a sensitivity analysis was conducted for the three main scaling parameters alpha, beta, and gamma. For each parameter, a range of values was evaluated while keeping all other parameters fixed at their default values. All evaluations were performed at noise level

$\sigma = 25$, where parameter sensitivity is expected to be highest.

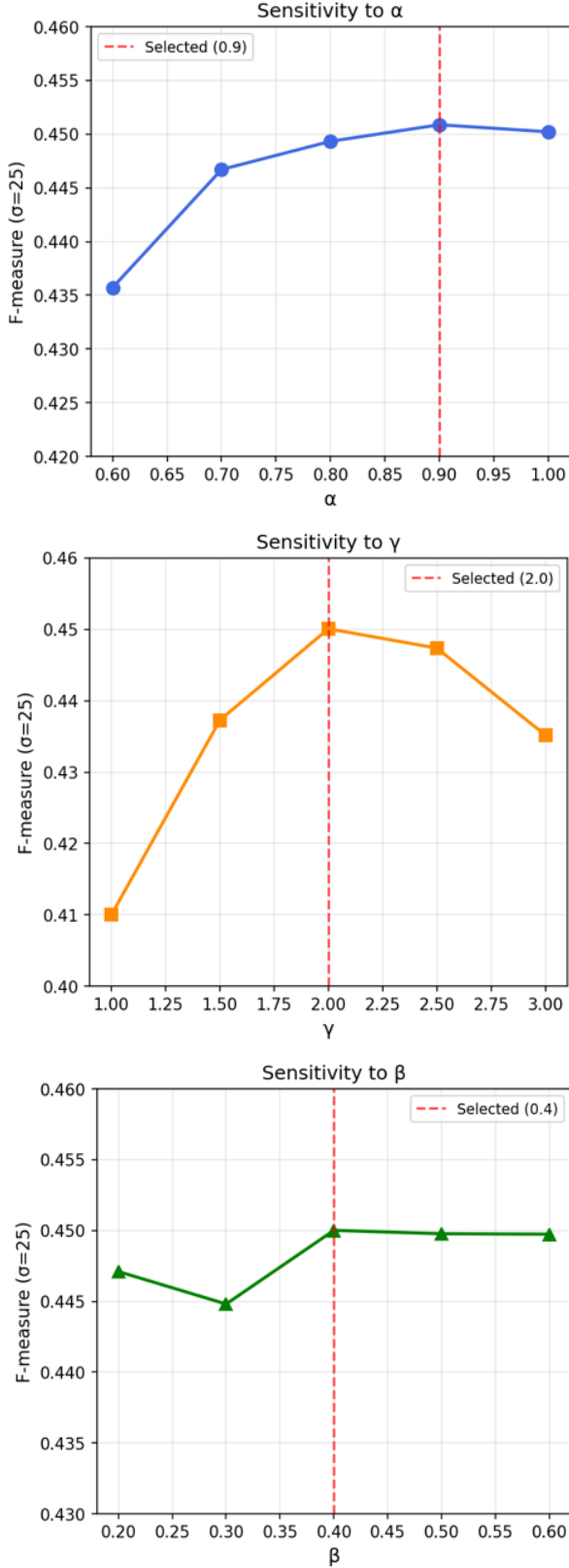


Fig. 4. Hyperparameter sensitivity analysis for α , γ , and β parameters at $\sigma = 25$.

Fig. 4 shows the F-measure as a function of each parameter. For alpha, performance increases steadily from 0.4357 at alpha = 0.6 to a peak of 0.4509 at alpha = 0.9, after which it plateaus. This suggests that alpha = 0.9 provides the

best balance between threshold sensitivity and noise suppression. For gamma, the optimal value is clearly gamma = 2.0, which corresponds to approximately two standard deviations of noise-induced gradient magnitude — a principled choice that aligns with the theoretical 95th percentile of a Gaussian distribution. Values below 2.0 underestimate the noise contribution, leading to thresholds that are too low, while values above 2.0 suppress too many true edges. For beta, performance is relatively stable across the tested range (0.2 to 0.6), with a slight optimum at beta = 0.4. This stability confirms that the high-to-low threshold ratio is not a critical tuning factor, and the default value follows established practice from the original Canny formulation.

Overall, the sensitivity analysis demonstrates that NCC is not highly sensitive to exact parameter values, particularly for beta. The most impactful parameter is gamma, which directly controls how aggressively the method compensates for noise in the threshold formula. The selected values produce near-optimal or optimal performance across all three sensitivity curves, confirming that the configuration used in the main experiments is well-justified.

E. Runtime Analysis

Average runtime per image: Sobel 11.64 ms, Standard Canny 23.48 ms, Otsu-Canny 39.78 ms, NCC 49.57 ms. The additional overhead of NCC over Standard Canny is approximately 26 ms, which is acceptable for the performance gain achieved.

The complete implementation code, including all baseline methods, the NCC pipeline, benchmark scripts, and result reproduction instructions, is publicly available at:

<https://github.com/uteodon/comp430-ncc-edge-detection/tree/main>

VI. DISCUSSION

The experimental results reveal several important insights about the behavior of edge detectors under noise. Standard Canny and Otsu-Canny both suffer from a fundamental limitation: their fixed smoothing parameter fails to suppress noise-induced gradient responses at high noise levels, resulting in very high recall but extremely low precision. This behavior indicates that these methods detect almost every pixel as an edge at $\sigma=25$, making their output practically unusable without post-processing.

Sobel performs competitively at low noise levels, achieving the highest F-measure of 0.5436 at $\sigma=5$. However, it lacks a principled thresholding mechanism, and its performance degrades significantly as noise increases. This suggests that gradient magnitude alone, without adaptive thresholding, is insufficient for robust edge detection across varying conditions.

The precision-recall analysis shown in Fig. 5 further illustrates the advantage of NCC. At $\sigma=25$, NCC occupies the upper-left region of the precision-recall space, indicating a well-balanced trade-off between detecting true edges and suppressing false detections. All other methods shift toward the lower-right region, reflecting high recall at the cost of very low precision.

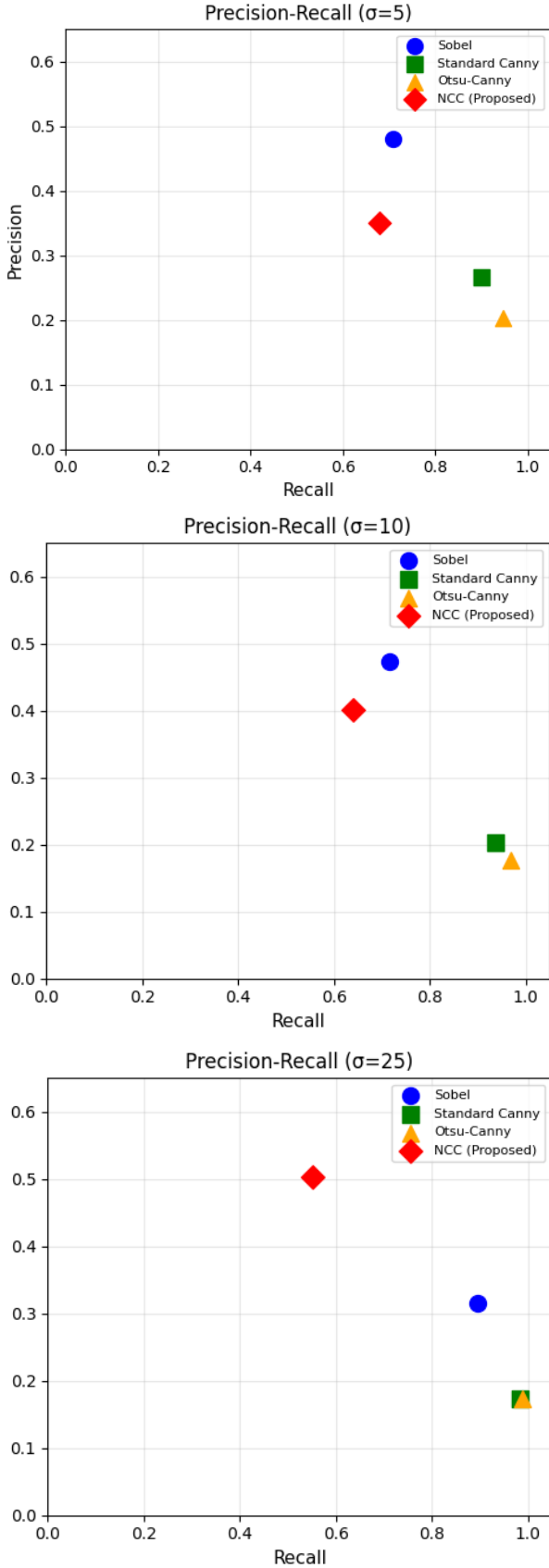


Fig. 5. Precision-Recall comparison of all methods across noise levels ($\sigma \in \{5, 10, 25\}$).

The ablation study confirms that neither adaptive smoothing alone nor adaptive thresholding alone is sufficient. NCC-A achieves better results than NCC-B across all noise levels, suggesting that noise-adaptive smoothing contributes

more than adaptive thresholding in isolation. However, NCC-Full consistently outperforms both variants, demonstrating that the two components are synergistic and must be applied together for optimal performance.

One limitation of the proposed method is its assumption of spatially uniform Gaussian noise. In real-world scenarios, noise may be spatially varying or follow non-Gaussian distributions. Additionally, the parameter values α , β , γ , and k were selected based on a fixed configuration rather than a systematic grid search, leaving room for further optimization. Future work will address these limitations by incorporating validation-based parameter tuning and extending the framework to handle structured and impulse noise.

It is also worth comparing the behavior of NCC against Otsu-Canny more carefully, since both methods aim to automate threshold selection. Otsu-Canny applies a statistical criterion to the gradient histogram without any knowledge of the underlying noise level. At low noise ($\sigma=5$), the gradient histogram retains a roughly bimodal structure, and Otsu's criterion can identify a reasonable separation between edge and non-edge gradients. However, as noise increases, the histogram becomes increasingly unimodal, and Otsu's threshold collapses toward lower values, admitting more noise responses. This explains why Otsu-Canny's precision drops from 0.2025 at $\sigma=5$ to 0.1726 at $\sigma=25$, while its recall simultaneously approaches 1.0. NCC avoids this failure mode entirely by anchoring T_{high} to both the gradient scale and the estimated noise level, making it robust to histogram shape changes.

The parameter sensitivity of NCC deserves further discussion. The four parameters α , β , γ , and k were fixed at 0.8, 0.4, 2.0, and 0.5 respectively based on preliminary analysis. The value $\gamma=2.0$ was chosen to approximate the 95th percentile of noise-induced gradient magnitudes under Gaussian noise, providing a principled upper bound on the noise contribution to T_{high} . The ratio $\beta=0.4$ follows Canny's original recommendation of a 1:2 to 1:3 ratio between low and high thresholds. While these values were not exhaustively optimized, the consistent improvement of NCC-Full over all baselines across all noise levels suggests that the method is not highly sensitive to exact parameter values, and that the noise-proportional structure of the threshold rule is the primary driver of performance gains.

It is also instructive to consider the failure cases of NCC. At very low noise levels ($\sigma=5$), NCC achieves a lower F-measure than Sobel (0.4475 vs 0.5436). This is because at low noise, σ_{est} is small, causing T_{high} to remain close to $\alpha \cdot \mu_{\text{grad}}$. In this regime, NCC behaves similarly to a fixed-threshold Canny detector, and Sobel's simpler gradient-based approach benefits from its broader edge response. This suggests that NCC is particularly well-suited for moderate to high noise conditions, while a hybrid strategy that switches between Sobel and NCC based on the estimated noise level could further improve performance across the full noise range.

Another consideration is the choice of ground truth representation. In this study, the union of all BSDS500 annotator boundaries was used as ground truth, producing a dense reference map. An alternative approach is to use the majority-vote boundary map, which retains only edges marked by more than half of the annotators. This stricter ground truth would likely increase precision scores across all methods while reducing recall, potentially changing the

relative ranking. Future work should evaluate NCC under both ground truth conventions to provide a more complete picture of its performance characteristics.

VII. CONCLUSION

This paper proposed the Noise-Calibrated Canny (NCC) framework, a lightweight and automatic parameter selection strategy for the Canny edge detector. By estimating the image noise level using a robust MAD-based statistic and deriving both the Gaussian smoothing scale and the hysteresis thresholds from this single estimate, NCC eliminates the need for manual parameter tuning. Experimental results on BSDS500 demonstrate that NCC is the only evaluated method whose F-measure improves as noise increases, achieving 0.5081 at $\sigma=25$ compared to 0.2906, 0.2891, and 0.4454 for Standard Canny, Otsu-Canny, and Sobel respectively. The ablation study confirms that adaptive smoothing and adaptive thresholding are complementary components, both necessary for robust performance.

The main limitation of the current method is that it assumes additive white Gaussian noise. Extension to impulse noise or structured noise would require additional preprocessing steps. Future work may also explore anisotropic diffusion as a preprocessing alternative to Gaussian smoothing for better edge preservation, and extension of the framework to color images.

Despite these limitations, the NCC framework demonstrates that substantial robustness gains can be achieved through principled noise estimation and parameter propagation, without the computational cost of deep learning methods or the manual effort of per-image parameter tuning. The simplicity and interpretability of the approach make it particularly well-suited for deployment in resource-constrained environments and as a preprocessing component in larger automated image analysis systems.

In summary, this work contributes a concrete, reproducible methodology for noise-aware edge detection,

validated on a standard benchmark with multiple noise levels and a rigorous ablation study. The results provide a strong baseline for future work on automatic parameter selection in classical edge detection pipelines.

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